SMART WASTE SORTING BIN

A3 DRAWINGS PACK · 智能分類垃圾桶 A3 圖紙

Version 1.0 · 29 October 2026 · Dimensions in mm

Contents

Fig. 01 System architecture · 系統架構

Fig. 02 Front section & sensor locations · 正視剖面與感測器位置

Fig. 03 Top view: bins, spout & sensors · 俯視圖

Fig. 04 Exploded assembly · 爆炸圖

Fig. 05 Detail A: weigh hopper, door & sensors · 詳圖 A

Fig. 06 One sort cycle · 一次分類流程

Fig. 07 Power distribution & safety wiring · 電源與安全接線

Fig. 08 ESP32 connection map · ESP32 接線圖

Fig. 09 Control state machine · 控制狀態機

Print at A3 (420 × 297 mm), landscape. Do not scale to fit A4; dimensions become unreadable.

The text tables in the design package are the authoritative build data. Verify all dimensions on the built frame before drilling.

System architecture

系統架構：two processors, one controlled waste path

Fig. 01

USER & CLOUD

REAL-TIME CONTROL

PHYSICAL PROCESS

Phone / PWA
Dashboard · history · manual sort
Wi-Fi hotspot or home network
Cloud sync (optional)
Firebase / Supabase · offline queue

Wi-Fi / REST

USB camera (UVC)

SENSOR SET

- **Camera** — material / shape
- **ToF item** — presence / height
- **Load cells** — weight
- **Moisture pad** — wet / dry
- **Metal coil** — metal confirm
- **IR beam** — hand safety
- **ToF ×4** — bin fill level
- **Switches** — lid / door / drawer

I2C / GPIO

Raspberry Pi Zero 2 W

VISION + APP + DATA

- **USB camera** — 5 MP UVC · 1280×720 · fixed exposure
- **Capture & pre-process** — 3 frames · crop · white balance
- **MobileNetV2 classifier** — TFLite · 4 classes + confidence
- **Sensor fusion** — vision + weight + moisture + metal
- **Sort orchestration** — state machine · retries · logging
- **FastAPI + SQLite** — REST API · event history · bin levels
- **Cloud sync** — optional · retry queue

USB serial

JSON

ESP32 DevKit (real-time MCU)

SENSORS + MOTION + SAFETY

- **Sensor acquisition** — HX711 · ToF ×5 · IR · moisture · metal
- **Debounce & filtering** — median / moving average / thresholds
- **Motion control** — stepper 3:1 belt · servo door · homing
- **Interlocks** — lid · drawer · door · e-stop · watchdog
- **Serial protocol** — newline JSON over USB · 115200 baud
- **Fault handling** — jam · timeout · bin missing · bin full
- **Status outputs** — RGB bar · buzzer · event log

motion

events

WASTE PATH

1	**Load**	lid open · item placed
2	**Weigh & see**	door closed · camera + sensors
3	**Decide**	4-class label + confidence
4	**Rotate spout**	stepper to selected bin
5	**Drop**	door opens · item falls
6	**Confirm & home**	drop beam · close · home

PWM / STEP

ACTUATORS

- **NEMA 17** — rotates spout — 3:1 GT2
- **Servo 25 kg** — hopper door — spring return
- **LED ring** — diffuse light — 12 V
- **RGB bar** — status — 5 V

POWER

12 V 6 Ah Li-ion (BMS) → 5 A fuse → main switch → e-stop → actuator bus

Buck 1: 5 V 5 A → Pi · Buck 2: 6 V 5 A → servo · Buck 3: 5 V 3 A → ESP32 / sensors

Common star ground · actuators cut by e-stop · Pi/ESP32 stay alive for logging

Signal types: USB / network · sensor bus · actuator power / PWM · power

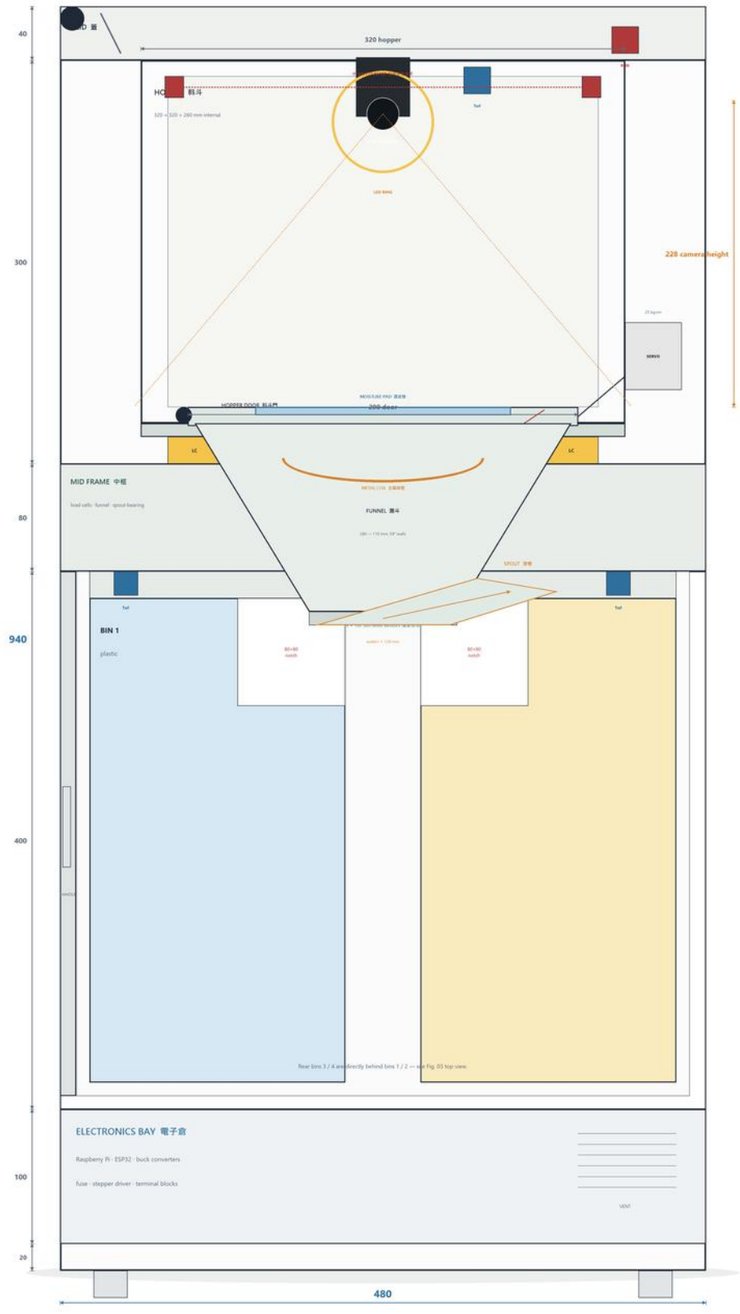
Two processors are used deliberately: Linux for vision/app, ESP32 for deterministic timing and safety.

Fig. 01 System architecture

Sensors and actuators around the ESP32; vision/app/data on the Raspberry Pi.

Front section & sensor locations
正視前面圖與感測器位置（mm）

Fig. 02



KEY COMPONENTS

1. **Lid with USB camera + diffuse LED ring**
   蓋：USB 鏡頭 + 擴散 LED 環
2. **IR safety beam across the hopper mouth**
   料斗口紅外安全光束
3. **Item ToF: presence & height**
   物件 ToF：存在與高度
4. **Weigh hopper on 4 × 5 kg load cells**
   4 × 5 kg 稱重感測器稱重料斗
5. **Capacitive moisture pad on door**
   門上電容式濕度點
6. **Fixed metal-detector coil under door**
   門下固定金屬探測線圈
7. **Servo + spring-assisted hopper door**
   伺服 + 彈簧輔助料斗門
8. **Funnel: 280 mm inlet, 110 mm outlet**
   漏斗：280 mm 入口，110 mm 出口
9. **Rotating spout on 150 mm bearing**
   150 mm 軸承旋轉滑槽
10. **4 fixed bins in a pull-out drawer**
    4 個固定桶於拉出抽屜
11. **4 × ToF bin-level sensors**
    4 × ToF 滿度感測器
12. **Electronics bay: Pi, ESP32, power**
    電子倉：Pi、ESP32、電源

DESIGN NOTES

- Overall 480 W × 480 D × 940 H mm; loading lip 900 mm.
- Hopper 320 × 320 × 260 mm internal; door 290 × 240 mm.
- Bins 190 × 190 × 360 mm internal, 80 × 80 mm corner notch.
- Usable bin volume = 10.7 L each; total = 42.8 L.
- Camera 90° diagonal FOV at 228 mm above door: 368 × 276 mm.
- Spout outlet radius 130 mm; bin centres at (±117, ±117).
- Nominal dimensions: leave 1–2 mm clearance and slotted mounts.

MATERIALS & FINISH

18 mm plywood frame; 3 mm PP liner in hopper and bins;
matte black inside hopper; silicone gasket on door;
stainless hinges; all wet surfaces removable for cleaning.

100 mm

Section A-A, viewed from the front. Schematic; verify on the built frame before drilling.

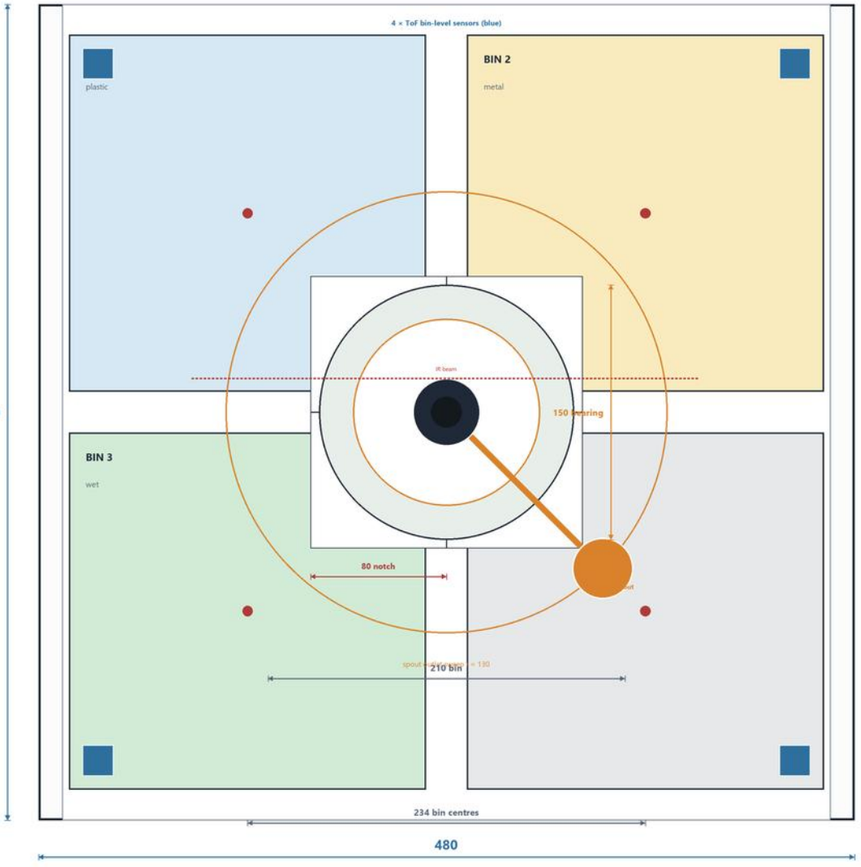
Fig. 02 Front section & sensor locations

Section A-A; nominal dimensions in mm.

Top view: bins, spout & sensor positions

俯視圖：垃圾桶、滑槽與感測器位置（mm）

Fig. 03



LAYOUT RULES

1. **Four fixed bins**
   2 × 2 grid; each 210 × 210 mm outer, 190 × 190 mm internal.
2. **Corner notches**
   80 × 80 mm at each inner corner clears the 150 mm bearing and funnel.
3. **Spout**
   Rotates 45°/135°/225°/315°; outlet radius 130 mm; always approaches clockwise.
4. **Bin centres**
   ±117 mm, ±117 mm; drop point is 35 mm inside each bin centre.
5. **Drawer**
   Slides out 400 mm; drawer-closed switch prevents sorting when open.
6. **Service**
   Lift bins out; wipe liner; check ToF windows and spout for jams.

BIN COLOUR KEY

- Plastic 塑料
- Metal 金屬
- Wet 濕垃圾
- Other 其他

Top view with lid removed. The camera and item ToF sit on the lid, centred over the hopper.

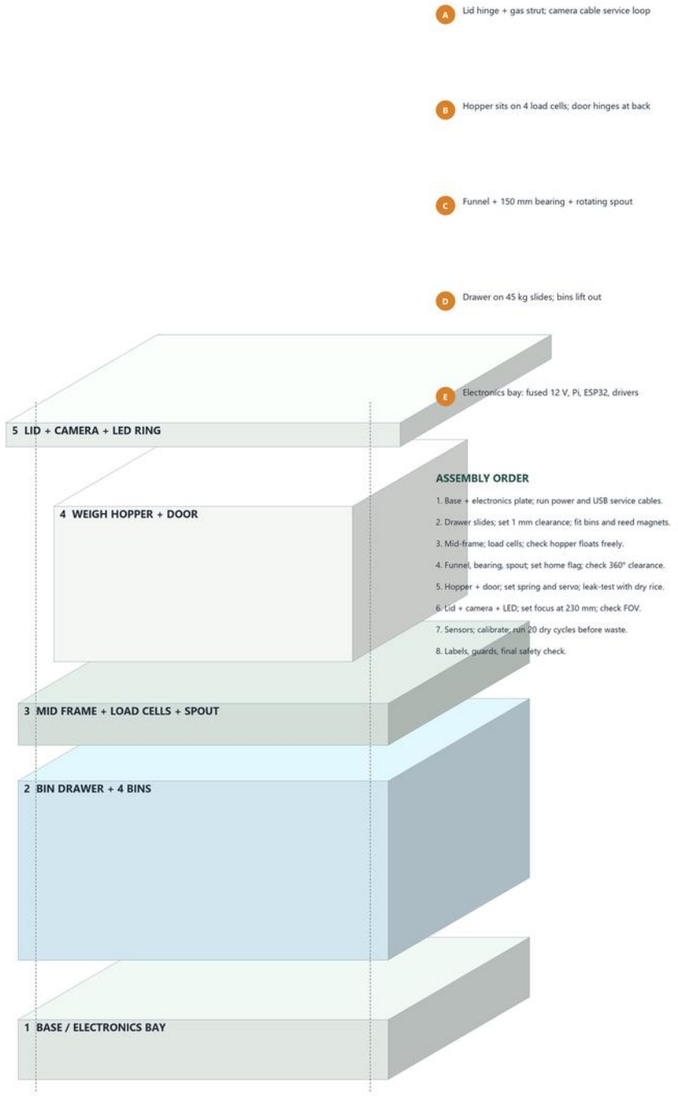
Fig. 03 Top view: bins, spout & sensors

2 × 2 bins, 80 × 80 mm corner notches, 150 mm bearing.

OVERALL 480 × 480 × 940 mm

Empty mass = 22-26 kg; four bins = 10.7 L each.

Built from 18 mm plywood + 3 mm PP liners; no welded frame.



ASSEMBLY ORDER

1. Base + electronics plate; run power and USB service cables.
2. Drawer slides; set 1 mm clearance; fit bins and reed magnets.
3. Mid-frame; load cells; check hopper floats freely.
4. Funnel, bearing, spout; set home flag; check 360° clearance.
5. Hopper + door; set spring and servo; leak-test with dry rice.
6. Lid + camera + LED; set focus at 230 mm; check FOV.
7. Sensors; calibrate; run 20 dry cycles before waste.
8. Labels, guards, final safety check.

Exploded view is schematic. Build the drawer and mid-frame first; test every subsystem before final assembly.

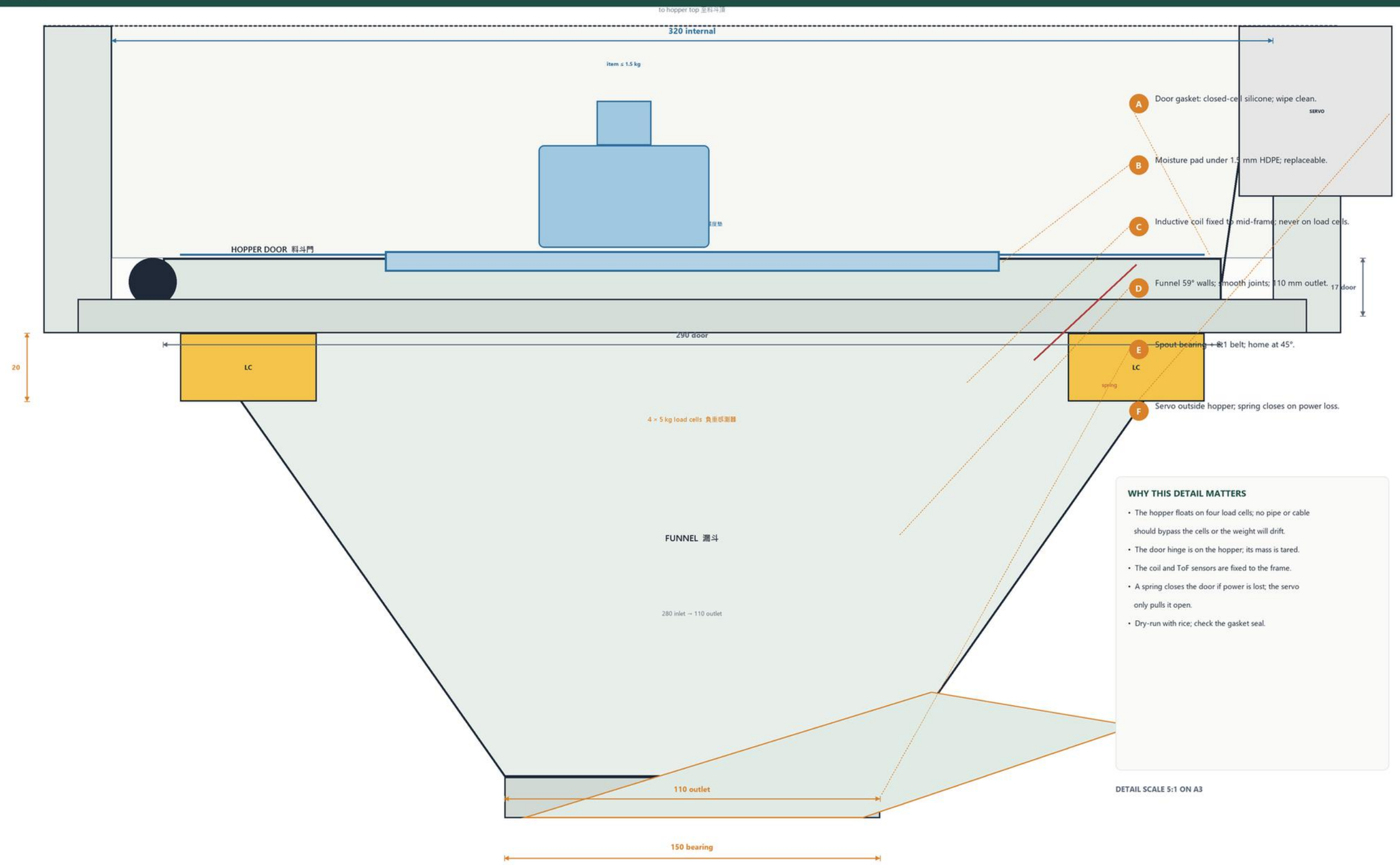
Fig. 04 Exploded assembly

Build from the base up; test the drawer, hopper and spout before final assembly.

Detail A: weigh hopper, door & sensors

詳圖 A：稱重料斗、門與感測器（mm）

Fig. 05



WHY THIS DETAIL MATTERS

- The hopper floats on four load cells; no pipe or cable should bypass the cells or the weight will drift.
- The door hinge is on the hopper; its mass is tared.
- The coil and ToF sensors are fixed to the frame.
- A spring closes the door if power is lost; the servo only pulls it open.
- Dry-run with rice; check the gasket seal.

DETAIL SCALE 5:1 ON A3

Detail A scale approximately 5:1 at A3. Confirm fit on the built hopper before cutting the liner.

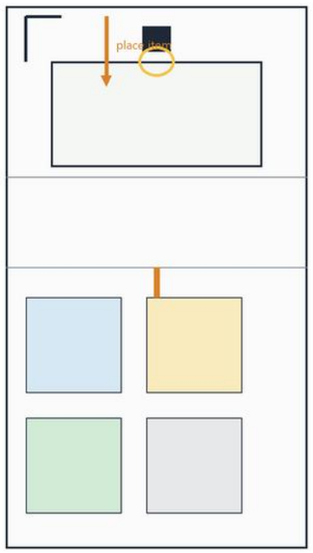
Fig. 05 Detail A: weigh hopper, door & sensors

Load cells, door, moisture pad, metal coil, funnel and bearing.

One sort cycle: 6-9 seconds
一次分類流程：6-9 秒

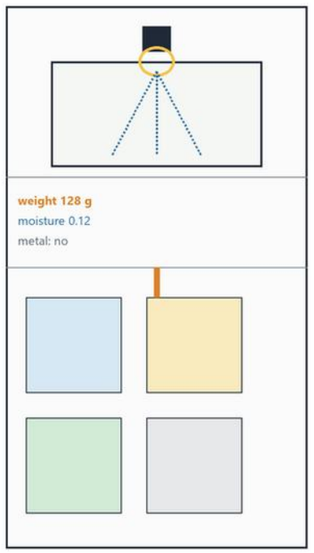
Fig. 06

1 LOAD
0-2 s



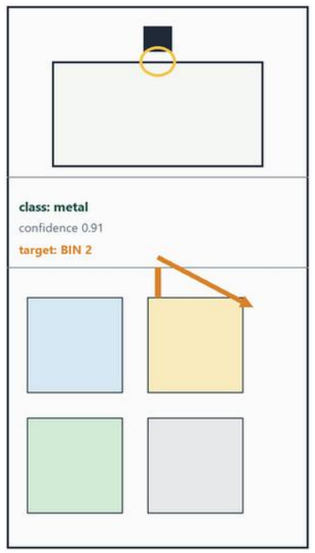
Lid closes; item sits on the hopper door.

2 SENSE
2-3 s



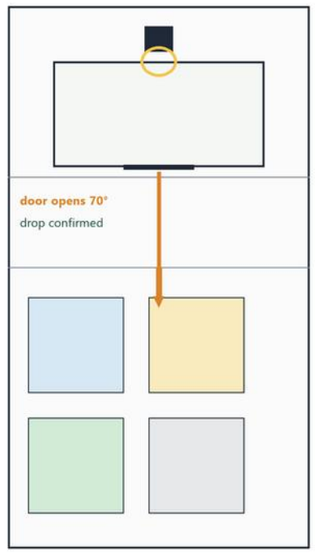
Load cells tare; camera captures 3 frames.

3 DECIDE + ROTATE
3-4 s



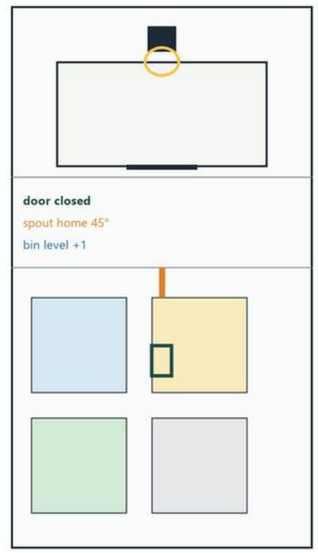
Pi classifies; ESP32 rotates spout to bin.

4 DROP
4-5 s



Door opens; item falls through funnel and spout.

5 CONFIRM + HOME
5-7 s



Drop beam confirms; close, home, log, update app.

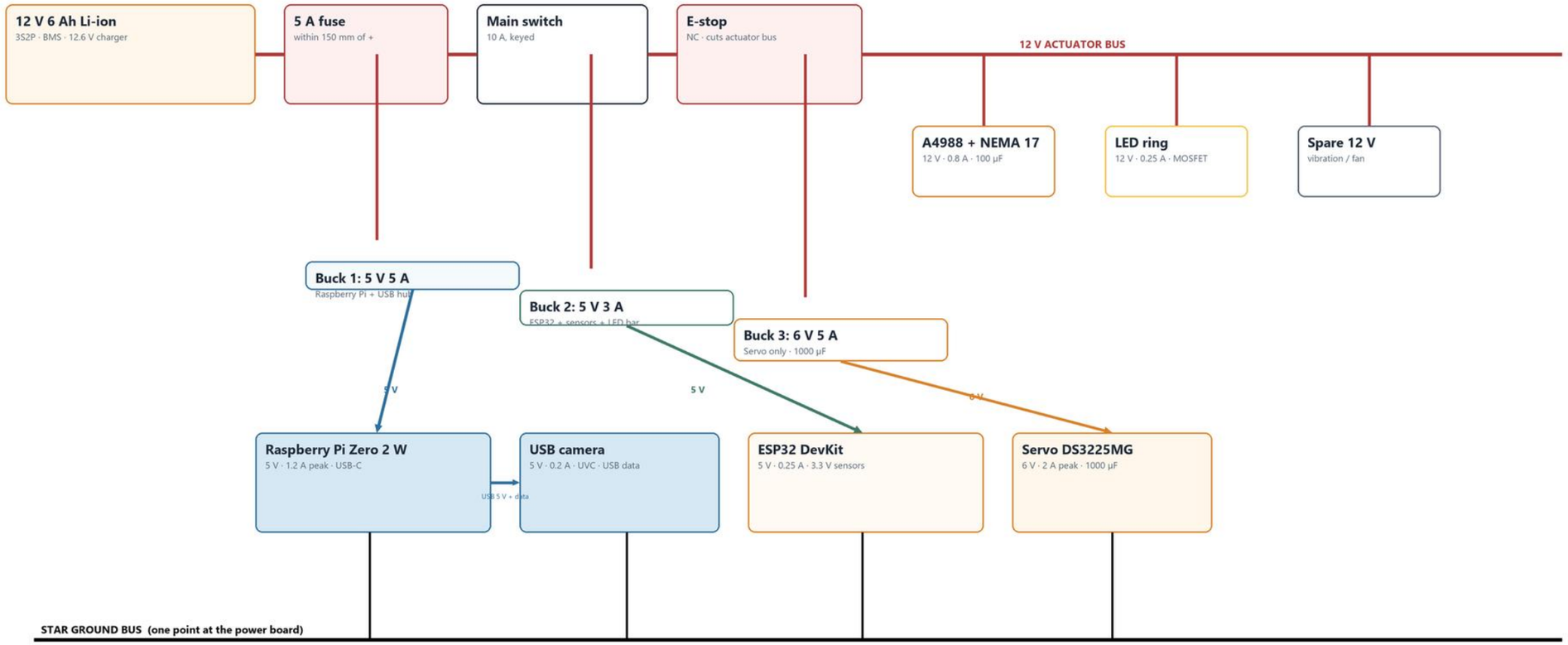
Timing excludes user loading time. The critical path is classification + spout rotation.

Fig. 06 One sort cycle: 6–9 seconds

Load, sense, decide/rotate, drop, confirm/home.

Power distribution & safety wiring
電源分配與安全接線

Fig. 07



WIRING RULES

- Fuse the battery positive within 150 mm of the pack.
- The e-stop cuts 12 V to stepper, servo and LED; Pi/ESP32 stay on 5 V.
- Use separate bucks for Pi, servo and ESP32; do not daisy-chain.
- Servo supply: 1000 µF electrolytic + 100 nF ceramic at the connector.
- A4988: 100 µF across VMOT/GND; set Vref = 0.64 V (0.8 A).
- Load-cell cable is shielded; shield grounded at HX711 only.
- Keep stepper and servo cables away from HX711 and I2C; cross at 90°.
- All connectors keyed and labelled; strain-relieve every cable.

WIRE GAUGE / COLOUR

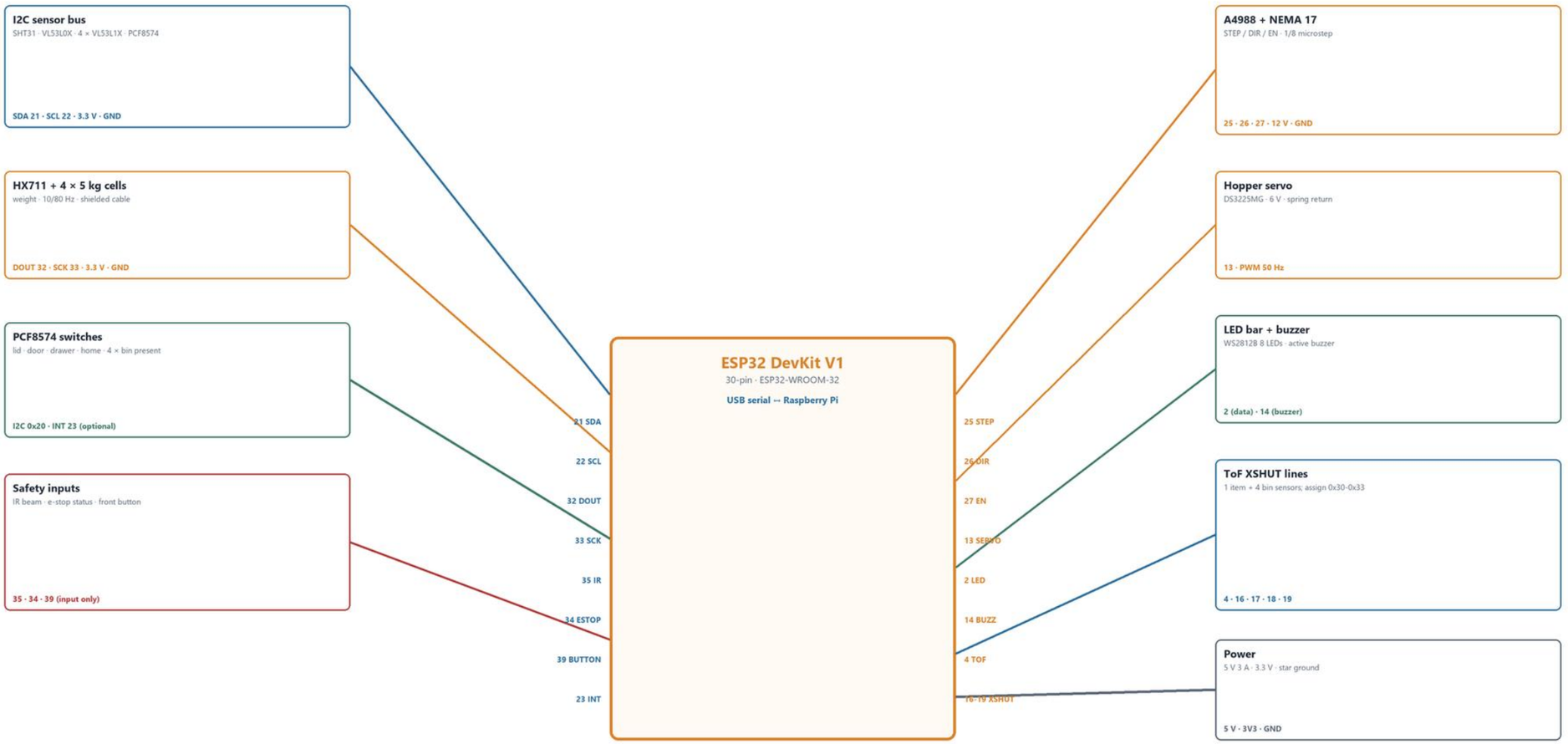
12 V battery / motor	18 AWG red / black	5 A fuse
5 V Pi / ESP32	20 AWG red / black	3 A buck
6 V servo	20 AWG orange / black	5 A buck
Signal / I2C	22-24 AWG twisted pair	keep < 500 mm
Load cell	4-core shielded	shield one end
Earth / frame	18 AWG green	star point

SAFETY Never charge a Li-ion pack unattended inside the bin. Use a certified pack with BMS; charge outside the enclosure.
The prototype uses 12 V SELV only. Do not route mains into the bin. Keep liquids away from the electronics bay.
Fit a 5 A fuse, a main switch and a mushroom e-stop. Label the e-stop "STOP MOTION / 停止動作".

Block diagram, not a full schematic. The pin-level connection map is Fig. 08; the terminal schedule is in Section 5.7.

Fig. 07 Power distribution & safety wiring

12 V SELV; e-stop cuts the actuator bus; Pi/ESP32 stay alive for logging.



PIN NOTES

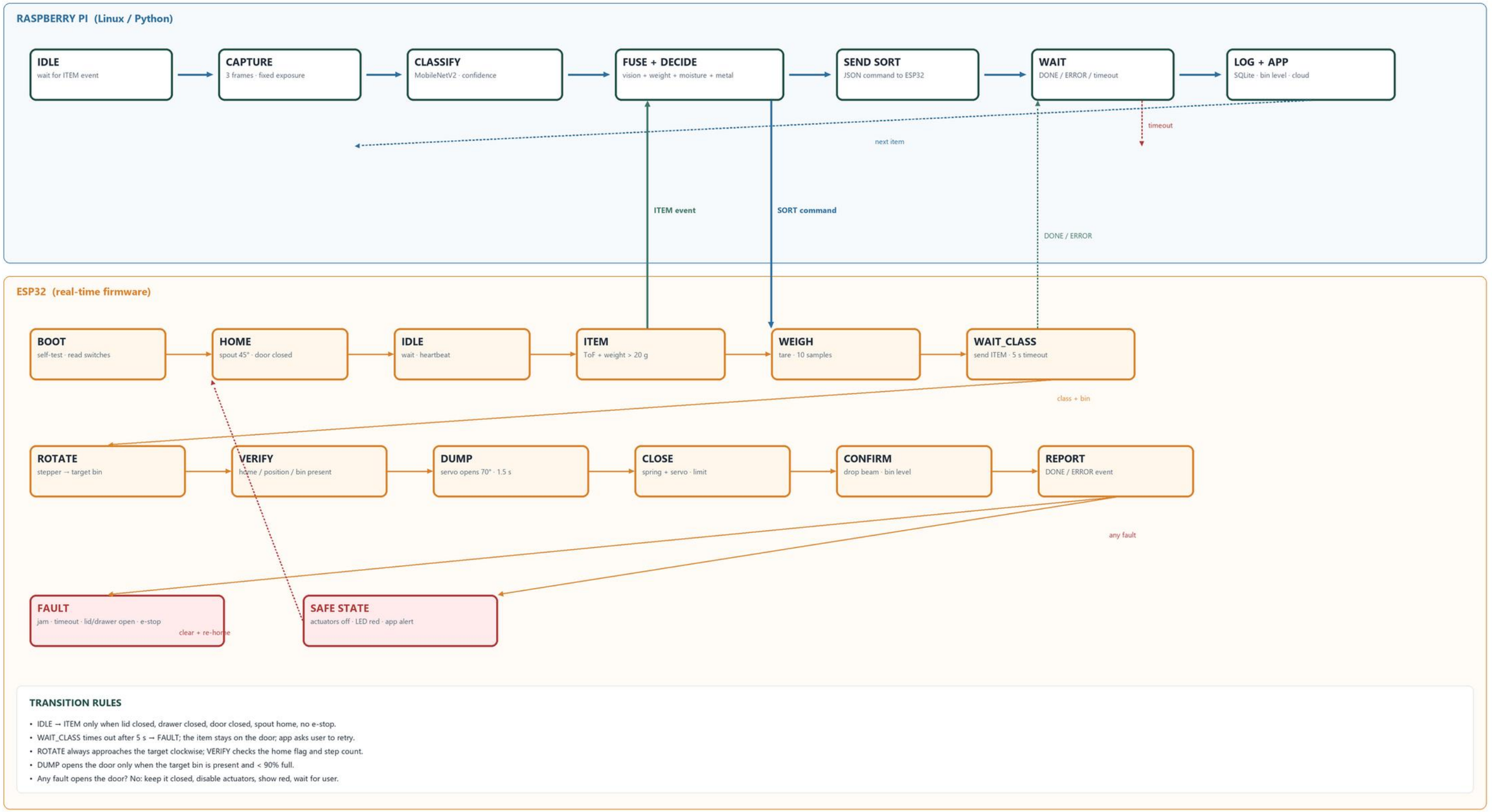
- GPIO34-39 are input-only: use external 10 kΩ pull-ups; no internal pull-up.
- GPIO2 is a strapping pin; it drives the LED data line only after boot. Do not hold it low at reset.
- All five ToF sensors share I2C; hold XSHUT low, release one at a time, and assign 0x30-0x33.
- A4988 Vref = 0.64 V for 0.8 A with Rs = 0.1 Ω. Use 1/8 microstep: 1200 steps per 90°.
- Servo power is 6 V, not 5 V; add 1000 µF at the servo connector and common ground.
- The Pi connects by USB; the ESP32 appears as /dev/ttyUSB0. Do not also connect 5 V from the Pi to the ESP32 5 V pin.

Fig. 08 ESP32 connection map

Logical map; use the pin table in Section 5.3 as the authoritative schedule.

Control state machine
控制狀態機

Fig. 09



The Pi can fail without causing motion: the ESP32 only moves on an explicit, validated SORT command.

Fig. 09 Control state machine

The Pi decides; the ESP32 executes only after validating interlocks and a target bin.